

BT4222 Project Deliverable Guideline

Academic Year 2026–2027 Term 1

Policy on Generative AI & AI Agents

Using AI tools to substitute for your own analytical thinking is strictly prohibited. You are encouraged to leverage LLMs and AI agents collaboratively as an intellectual thought partner, debugger, and learning multiplier across the following capacities:

- **Conceptual Sparring:** Discussing problem formulations, exploring alternative modeling logic, and identifying potential blind spots or edge cases in your system design.
- **Critical Evaluation & Debugging:** Identifying errors, invalid assumptions, or hallucinations in AI-generated code, data processing pipelines, and metric calculations.
- **Expanded Technical Learning:** Leveraging AI to accelerate your understanding of relevant literature, explore advanced modeling techniques, and refine your implementation workflows.
- **Deliverable Requirement:** Include an **AI Interaction Log** in your report appendix detailing essential prompts, AI responses, your critical evaluations, and the subsequent revisions made (remove routine or irrelevant prompts; do not include all of the scripts generated by AI).

Deliverable List

Ensure all project source code and datasets are accessible under a single GitHub repository.

Deliverable	Format	Due Time
Project Final Presentation Slides	.pptx uploaded to Canvas (preloaded onto the presentation laptop)	11:59 PM Nov. 10 (Tue.)
Project Final Report	.pdf or .docx uploaded to Canvas	11:59 PM Nov. 10 (Tue.)
Project Source Code	.ipynb / .py scripts uploaded to GitHub	11:59 PM Nov. 09 (Mon.)
Datasets & Preprocessing Artifacts	Cleaned/intermediary .csv or accessible repository links	11:59 PM Nov. 09 (Mon.)
Group Peer Evaluation Form	.doc or .docx uploaded to Canvas	11:59 PM Nov. 11 (Wed.)

Note: Submit a single PDF to Canvas by **11:59 PM Nov. 09 (Mon.)** containing all verified GitHub repository links and short descriptions of the file structures.

Requirement for Source Code

- **Self-Explanatory Pipeline:** Organize files systematically by stage (e.g., step1_data_preprocess.ipynb, step2_baselines.ipynb, step3_advanced_models.ipynb).
- **Execution & Reproducibility:** Code must run directly in Google Colab / GPU environments, with ALL performance output cells preserved and highlighted.
- **Documented Pipeline:** Use clear markdown documentation within notebooks to explain the purpose, approach, and logic of important functions and modeling steps.

Format of the Project Final Report

- **Cover Page:** Project Topic, Group Number, List of Group Members.
- **Formatting:** Line spacing 1.5 lines; Font: Arial, Calibri, or Times New Roman, Size 11; Normal Margins, Single Column.

Content of the Project Final Report

Abstract (Standalone page, within 300 words)

- **Business Objective & RecSys Formulation:** Define the core recommendation problem grounded in an exploratory understanding of dataset distributions and domain constraints.
- **Prior Baseline & Critical Diagnosis:** Frame your starting point with reference to past project baselines (e.g., NCF, NMF), identifying their structural limitations, methodological gaps, or performance bottlenecks.
- **Group Contributions & Modeling Highlights:** Summarize your group's engineering, architectural, and analytical improvements over the baseline.
- **Evidence-Backed Quantitative Insights:** State key business and behavioral insights supported directly by concrete quantitative metrics, offline evaluation, or diagnostic error analysis.

Section 1: Data Understanding & Feature Engineering

Instead of repeating the interim report, focus on updating changes, data distributions, and incremental features developed (e.g., utilizing impression feedback, interaction types, or context signals). Use structured tables where appropriate.

Section 2: Models and Performance (Within 4 pages)

- **Architecture Diagrams:** Provide clear, modular diagrams illustrating model architecture, input pipelines, and data flow.
- **Performance Reporting:** Report both training and testing performance across baseline and advanced models using appropriate ranking metrics with clear explanations of sample splits.

Section 3: Contribution and Justification (1–3 pages per chosen aspect)

Complete the Self-Evaluation Table below. Groups are not required to cover all optional aspects; focus on the areas matching your team's technical depth.

Self-Evaluation Table

Contribution Aspect	Dimension	Basic	Optional / Advanced
1. Data Understanding, Rich Signal Utilization & Feature Engineering <i>(Impressions, multi-type user interactions, context, domain features)</i>	Effort		
	Effectiveness		
<i>Clarification:</i> • Basic: e.g. Convert data into standard binary interaction pairs without deeper exploration of additional information; apply standard transformations or descriptive statistics. • Optional: e.g. Thoroughly analyze data distributions and sparsity prior to modeling; exploit rich signals (e.g., impression logs, exposed vs. unexposed items, multi-behavior interactions, timestamps/sessions) and engineer domain-informed features to enrich predictive signals.			
2. Prior Baseline Benchmarking & Gap Diagnosis <i>(Critical evaluation of NCF, NMF, and conventional baselines)</i>	Effort		
	Effectiveness		
<i>Clarification:</i> • Basic: e.g. Replicate standard baseline models (e.g., NCF, NMF) directly to obtain comparative metrics. • Optional: e.g. Critically diagnose structural failure modes or information bottlenecks in baseline implementations; systematically justify your improvements against these identified gaps.			
3. Designing or Adapting Recommender Architectures	Effort		
	Effectiveness		
<i>Clarification:</i> • Basic: e.g. Direct implementation of standard models or simple ensembling. • Optional: e.g. Adapt or customize architectures capturing temporal dynamics, sequential interaction			

patterns, or network/relational structures to improve recommendation quality.			
4. Model Diagnostics, Error Analysis & Quantitative Insights	Effort		
	Effectiveness		
<i>Clarification:</i> • Basic: e.g. Describe aggregate test metrics without deeper sub-group analysis. • Optional: e.g. Perform granular error diagnostics across distinct user/item segments; conduct deep learning model interpretation by examining learned parameters (e.g., analyzing attention weights, etc.); extract actionable behavioral and business insights supported by empirical evidence.			
5. Smart Use of AI Tools & Critical Collaboration	Effort		
	Effectiveness		
<i>Clarification:</i> • Basic: Use AI tools strictly for routine coding syntax, formatting, or text polishing. • Optional: Engage AI as a critical sparring partner to explore modeling alternatives, catch conceptual/coding flaws, validate technical logic, and drive expanded learning documented in the AI Interaction Log.			

Section 4: AI Interaction Log (Appendix)

Provide a concise table documenting key collaborative interactions with AI tools:

1. **Core Prompt & Task Objective:** The specific question, modeling concept, or technical problem being explored.
2. **AI Output Summary:** The substantive response or code generated (do not include all of the code generated by AI in the Appendix).
3. **Critical Evaluation:** Your assessment of limitations, naive assumptions, or errors in the AI response.
4. **Refinement & Final Action:** How you validated, adjusted, and incorporated the solution into your project.

Section 5: References (Within 1 pages)

Comprehensive citations for all datasets, baseline repositories, and referenced academic literature.

Assessment Rubrics (42 Marks)

- **Deliverable Integrity & Reporting Standards (10 Marks):** On-time submission following required formatting; complete Abstract, Sections 1–2, and Sections 4–5 structured with clarity and factual rigor.
- **Source Code Quality, Reproducibility & Modularity (5 Marks):** Comply with the source code requirements. Clean, modular pipeline; reproducible baseline and advanced results; clear Colab/GPU execution instructions with output cells preserved.
- **Technical Contribution & Effectiveness across the Five Aspects (17 Marks):** Evaluated across the 5 dimensions in the Self-Evaluation Table (Data Understanding & Feature Engineering, Baseline Diagnosis, Architecture Design, Diagnostics & Insights, and Smart Use of AI Tools), assessing both the extent of efforts and effectiveness of outcomes.
- **Final Presentation Delivery & Mastery (10 Marks):**
 - **Basic Requirement:** 10 minutes per group excluding Q&A; every group member must present; reading scripts or slides throughout the presentation is prohibited.
 - **Presentation Quality & Technical Mastery:** Logical organization, clarity, and succinct communication of insights. The presentation must reflect the group's deep understanding rather than superficial recitation, demonstrating the ability of any member to fluently explain data preprocessing, model logic/diagnostics/analysis, and code implementation during the presentation and Q&A.

Group Peer Evaluation

To avoid disputes and receiving unexpected outcomes from peer evaluation:

- Define an agreeable team project policy in advance.
- Provide objective and specific justifications for any negative evaluations; subjective claims cannot be verified.
- *Note:* Individual Mark = (Group Interim Mark + Group Final Mark) * Average Group Peer Evaluation (%).